

Probability basics

Last modified: 2026-09-28 14:24:58 (PDT)

Defining probabilities

Definition 0.1 (Sample space). The **sample space** of a random experiment, denoted Ω , is the set of all of its possible outcomes.

Example 0.1 (Sample space of a die roll). Rolling a six-sided die once has sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$.

Definition 0.2 (σ -algebra). A **σ -algebra** on a set S is a collection $\mathcal{S}$ of subsets of S that satisfies:

- $\mathcal{S}$ contains S itself.
- For each set A in $\mathcal{S}$, $\mathcal{S}$ contains its complement $S \setminus A$.
- For each sequence $A_1, A_2, \dots$ of sets in $\mathcal{S}$, $\mathcal{S}$ contains their union $\bigcup_{i=1}^{\infty} A_i$.

Other sources call it a **σ -field** (see [Wikipedia: \$\sigma\$ -algebra](#)). A σ -algebra also contains $\emptyset = S \setminus S$, every finite union of its sets (pad the sequence with copies of $\emptyset$), and every countable intersection of its sets (take complements, apply the union rule, and take the complement again).

Example 0.2 (σ -algebras for a die roll). For the die roll in Example 0.1, the collection of all subsets of $\Omega = \{1, 2, 3, 4, 5, 6\}$ is a σ -algebra on Ω , since complements and unions of subsets of Ω are again subsets of Ω . So is the smaller collection

$$\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, \Omega\}$$

which contains Ω , the complement of each of its sets, and every union of its sets: for example, $\{2, 4, 6\} \cup \{1, 3, 5\} = \Omega$.

Definition 0.3 (Event). An **event** is a subset of the **sample space** Ω : a set of outcomes.

When Ω is uncountable, such as an interval of real numbers, only the subsets in a designated **σ -algebra** $\mathcal{S}$ on Ω count as events; every set that arises in these notes is in $\mathcal{S}$. When Ω is finite or countably infinite, every subset of Ω can be an event, as in Example 0.2.

Example 0.3 (Rolling an even number). In Example 0.1, the event “the roll is even” is $A = \{2, 4, 6\}$.

Definition 0.4 (Occurrence of an event). An **event** A **occurs** when the outcome of the experiment is in A , and does not occur when the outcome is not in A .

Example 0.4 (Rolling a 4). In Example 0.3, if the die shows 4, the event “the roll is even”, $\{2, 4, 6\}$, occurs, because $4 \in \{2, 4, 6\}$, and the event “the roll is at most 2”, $\{1, 2\}$, does not occur.

Definition 0.5 (Complement of an event). The **complement** of an **event** A , denoted $\neg A$, is the event that A does not **occur**: the set of outcomes in the sample space Ω that are not in A .

$$\neg A \stackrel{\text{def}}{=} \Omega \setminus A$$

Other sources write A^c or $\bar{A}$.

Example 0.5 (Rolling an odd number). In Example 0.3, the complement of the event “the roll is even”, $A = \{2, 4, 6\}$, is the event “the roll is odd”, $\neg A = \{1, 3, 5\}$.

Definition 0.6 (Mutually exclusive events). Finitely or countably many sets $A_1, A_2, \dots$, such as **events**, are **mutually exclusive** (also called **disjoint**, **pairwise disjoint**, or **mutually disjoint**) when no two of them share an element:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

Example 0.6 (Low and high rolls). In Example 0.1, the events “the roll is at most 2”, $\{1, 2\}$, and “the roll is at least 5”, $\{5, 6\}$, are mutually exclusive: no roll is in both. The events “the roll is even”, $\{2, 4, 6\}$, and “the roll is at most 2”, $\{1, 2\}$, are not mutually exclusive: a roll of 2 is in both.

Definition 0.7 (Partition of an event). A **partition** of an **event** A is a finite or countably infinite collection of events $A_1, A_2, \dots$ that satisfies:

- The events $A_1, A_2, \dots$ are **mutually exclusive**.
- Their union is A : $\bigcup_i A_i = A$.

A partition of the **sample space** is the case $A = \Omega$.

Some sources also require each piece A_i to contain at least one outcome.

Example 0.7 (Partitioning the die roll). In Example 0.1, the events “the roll is even”, $\{2, 4, 6\}$, and “the roll is odd”, $\{1, 3, 5\}$, are mutually exclusive, and their union is Ω , so they partition the sample space. So do the six single-outcome events $\{1\}, \{2\}, \dots, \{6\}$. The events $\{1, 2\}$ and $\{5, 6\}$ from Example 0.6 do not partition Ω : their union leaves out 3 and 4.

Definition 0.8 (Finite additivity). Let $\mathcal{S}$ be a **σ -algebra** on a set S . A function μ that assigns a value $\mu(A) \in [0, \infty]$ to each set A in $\mathcal{S}$ is **finitely additive** if, for every finite collection of **pairwise disjoint** sets $A_1, \dots, A_n$ in $\mathcal{S}$, the value of their union is the sum of their values:

$$\mu(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mu(A_i)$$

Example 0.8 (Counting outcomes is finitely additive). For the die roll in Example 0.1, let $\mu(A) \stackrel{\text{def}}{=} |A|$, the number of outcomes in A , for each set A in the σ -algebra of all subsets of Ω (Example 0.2). The events $\{1, 2\}$ and $\{5, 6\}$ are mutually exclusive (Example 0.6), and:

$$\begin{aligned} \mu(\{1, 2\} \cup \{5, 6\}) &= \mu(\{1, 2, 5, 6\}) && \text{(take the union)} \\ &= 4 && \text{(count the outcomes)} \\ &= 2 + 2 && \text{(write 4 as a sum)} \\ &= \mu(\{1, 2\}) + \mu(\{5, 6\}) && \text{(count each event's outcomes)} \end{aligned}$$

The same holds for any mutually exclusive events, because the sizes of disjoint sets add, so μ is finitely additive.

Definition 0.9 (Countable additivity). Let $\mathcal{S}$ be a **σ -algebra** on a set S . A function μ that assigns a value $\mu(A) \in [0, \infty]$ to each set A in $\mathcal{S}$ is **countably additive** (also called **σ -additive**) if, for every sequence of **pairwise disjoint** sets $A_1, A_2, \dots$ in $\mathcal{S}$, the value of

their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

A sum of values in $[0, \infty]$ is either a finite number or ∞ , so the right-hand side always has a value.

Example 0.9 (Counting outcomes is countably additive). For the counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.8, take any sequence of pairwise disjoint events $A_1, A_2, \dots$. No two of them share an outcome, and Ω has only six outcomes, so at most six of the A_i contain any outcomes; the rest are $\emptyset$, with $\mu(\emptyset) = 0$. The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.8) shows that those terms add up to μ of the union. So μ is countably additive.

Lemma 0.1 (Countable additivity and the empty set). *If μ is a **countably additive** function on a **σ -algebra** $\mathcal{S}$, then $\mu(\emptyset) = 0$ or $\mu(\emptyset) = \infty$.*

i Proof

Proof. The set $\emptyset = S \setminus S$ is in $\mathcal{S}$, as the complement of S , and the sequence $\emptyset, \emptyset, \dots$ is pairwise disjoint with union $\emptyset$, so:

$$\begin{aligned} \mu(\emptyset) &= \mu\left(\bigcup_{i=1}^{\infty} \emptyset\right) \quad (\text{the union of copies of } \emptyset \text{ is } \emptyset) \\ &= \sum_{i=1}^{\infty} \mu(\emptyset) \quad (\text{countable additivity}) \end{aligned}$$

If $\mu(\emptyset) = c$ for a finite $c > 0$, the right-hand side is $c + c + \dots = \infty \neq c$, a contradiction. So $\mu(\emptyset)$ is 0 or ∞ . $\square$

Both values occur. The counting function of Example 0.9 has $\mu(\emptyset) = 0$, while the function with $\mu(A) = \infty$ for every A , including $\emptyset$, is countably additive, since both sides of the defining equation are ∞ . So $\mu(\emptyset) = 0$ is an extra requirement, not a consequence of countable additivity.

Theorem 0.1 (Countable additivity implies finite additivity). *If μ is a **countably additive** function on a **σ -algebra** $\mathcal{S}$, and $\mu(\emptyset) = 0$, then μ is **finitely additive**.*

i Proof

Proof. Let $A_1, \dots, A_n$ be pairwise disjoint sets in $\mathcal{S}$, and extend them to a sequence by setting $A_{n+1} = A_{n+2} = \dots = \emptyset$. The extended sequence is still pairwise disjoint, since $\emptyset$ shares no element with any set, and its union is $A_1 \cup \dots \cup A_n$. So:

$$\begin{aligned}
 \mu(A_1 \cup \dots \cup A_n) &= \mu\left(\bigcup_{i=1}^{\infty} A_i\right) && (A_i = \emptyset \text{ for } i > n) \\
 &= \sum_{i=1}^{\infty} \mu(A_i) && (\text{countable additivity}) \\
 &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) && (A_i = \emptyset \text{ for } i > n) \\
 &= \sum_{i=1}^n \mu(A_i) && (\mu(\emptyset) = 0)
 \end{aligned}$$

□

The converse fails: there exist set functions that satisfy finite additivity but fail countable additivity (see [Wikipedia: Sigma-additive set function — An additive function which is not \$\sigma\$ -additive](#)).

Definition 0.10 (Measure). A **measure** on a set S with a σ -algebra $\mathcal{S}$ is a function μ that assigns a value $\mu(A) \in [0, \infty]$ to each set A in $\mathcal{S}$ and satisfies:

- $\mu(\emptyset) = 0$.
- μ is **countably additive**.

A measure is also **finitely additive**, by Theorem 0.1. A measure generalizes size: it can measure how many elements a set has, as in Definition 0.11, or how long a set of real numbers is, as [Lebesgue measure](#) does, assigning each interval $[a, b]$ its length $b - a$. Both appear as reference measures in the [joint-distribution form of Fubini–Tonelli](#).

Example 0.10 (Counting outcomes is a measure). The counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.8, defined on the σ -algebra of all subsets of the die's sample space (Example 0.2), takes values in $[0, \infty]$, is countably additive (Example 0.9), and gives $\mu(\emptyset) = 0$, so it is a measure.

Definition 0.11 (Counting measure). The **counting measure** on a set S is the [measure](#) on the σ -algebra of all subsets of S that assigns each finite set its number of elements, and each infinite set the value ∞ :

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

Example 0.11 (Counting measure on the non-negative integers). For the counting measure μ on $\{0, 1, 2, \dots\}$, $\mu(\{0, 1, 2\}) = 3$, and the set of even numbers has $\mu(\{0, 2, 4, \dots\}) = \infty$. The even numbers are the union of the pairwise disjoint sets $\{0\}, \{2\}, \{4\}, \dots$, and countable additivity agrees: $\sum_{k=0}^{\infty} \mu(\{2k\}) = 1 + 1 + \dots = \infty$.

Definition 0.12 (Probability measure). A **probability measure** on a **sample space** Ω , often denoted $\Pr()$ or $P()$, is a function that assigns a number $\Pr(A)$ to each **event** A and satisfies:

- $\Pr$ is a **measure** on the events of Ω .
- The whole sample space has probability 1: $\Pr(\Omega) = 1$.

Many sources state this definition as three axioms instead (the Kolmogorov axioms):

1. For any event A , $\Pr(A) \geq 0$.
2. The probability of the whole sample space is 1:

$$\Pr(\Omega) = 1$$

3. $\Pr$ is **countably additive**: for any **mutually exclusive** events $A_1, A_2, \dots$,

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

The two forms are equivalent. The axioms do not list $\Pr(\emptyset) = 0$, but they imply it: applying property 3 to $\Omega, \emptyset, \emptyset, \dots$ gives $1 = 1 + \sum_{i=2}^{\infty} \Pr(\emptyset)$, so $\Pr(\emptyset) = 0$. A probability measure is also **finitely additive**, by Theorem 0.1. Requiring countable additivity, not just finite additivity, enables results such as the continuity of probability (if $A_1 \supseteq A_2 \supseteq \dots$ with $\bigcap_i A_i = \emptyset$, then $\Pr(A_i) \rightarrow 0$), and it is what makes sums over countably infinite **partitions** valid.

Example 0.12 (Probability measure for a fair die). For the die roll in Example 0.1, define $\Pr(A) \stackrel{\text{def}}{=} |A|/6$, where $|A|$ is the number of outcomes in A . This function is $1/6$ times the measure of Example 0.10, so it is a **measure** on the events of Ω , and it gives $\Pr(\Omega) = 6/6 = 1$. The event “the roll is even” from Example 0.3 has probability $\Pr(\{2, 4, 6\}) = 3/6 = 1/2$.

Theorem 0.2 (Probability of a subset's intersection). *If A and B are events and $A \subseteq B$, then $\Pr(A \cap B) = \Pr(A)$.*

i Proof

Proof. Since $A \subseteq B$, every outcome in A is also in B , so $A \cap B = A$, and therefore $\Pr(A \cap B) = \Pr(A)$. $\square$

Theorem 0.3 (An event and its complement sum to 1). *For any event A and its complement $\neg A$:*

$$\Pr(A) + \Pr(\neg A) = 1$$

i Proof

Proof. The events A and $\neg A$ are disjoint, and their union is Ω .

$$\begin{aligned}\Pr(A) + \Pr(\neg A) &= \Pr(A \cup \neg A) && \text{(additivity of probability for disjoint events)} \\ &= \Pr(\Omega) && (A \cup \neg A = \Omega) \\ &= 1 && \text{(probability of the sample space is 1)}\end{aligned}$$

$\square$

Corollary 0.1 (Complement rule). *For any event A :*

$$\Pr(\neg A) = 1 - \Pr(A)$$

i Proof

Proof. Subtract $\Pr(A)$ from both sides of Theorem 0.3. $\square$

Corollary 0.2 (Complement rule in probability (π) notation). *If the probability of an event A is $\Pr(A) = \pi$, then the probability that A does not occur is:*

$$\Pr(\neg A) = 1 - \pi$$

i Proof

Proof.

$$\begin{aligned}\Pr(\neg A) &= 1 - \Pr(A) && \text{(complement rule)} \\ &= 1 - \pi && \text{(substitute } \Pr(A) = \pi\text{)}\end{aligned}$$

□

Example 0.13 (Probability of not rolling a six). For a fair die, the event “the roll is a six” has probability $\pi = 1/6$, so by Corollary 0.2 the probability of not rolling a six is $1 - 1/6 = 5/6$.

Conditional probability

Definition 0.13 (Conditional probability). For two events A and B with $\Pr(B) > 0$, the **conditional probability** of A given B , denoted $\Pr(A \mid B)$, is:

$$\Pr(A \mid B) \stackrel{\text{def}}{=} \frac{\Pr(A \cap B)}{\Pr(B)}$$

Example 0.14 (Rolling a six, given an even roll). For a fair die (Example 0.12), let $A = \{6\}$ and $B = \{2, 4, 6\}$. Then $A \cap B = \{6\}$, so:

$$\begin{aligned}\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\ &= \frac{1/6}{3/6} && \text{(substitute the probabilities)} \\ &= \frac{1}{3} && \text{(simplify)}\end{aligned}$$

Theorem 0.4 (Law of conditional probability). *For any two events A and B with $\Pr(B) > 0$:*

$$\Pr(A \cap B) = \Pr(A \mid B) \cdot \Pr(B)$$

i Proof

Proof. Rearranging Definition 0.13:

$$\Pr(A | B) = \frac{\Pr(A \cap B)}{\Pr(B)} \quad (\text{definition of conditional probability})$$

$$\Pr(A \cap B) = \Pr(A | B) \cdot \Pr(B) \quad (\text{multiply both sides by } \Pr(B))$$

□

Example 0.15 (Applying the law of conditional probability). Suppose 30% of adults exercise regularly ($\Pr(E) = 0.30$), and among adults who exercise regularly, 60% have low blood pressure ($\Pr(L | E) = 0.60$).

Then, by Theorem 0.4, the probability that a randomly selected adult both exercises regularly and has low blood pressure is:

$$\begin{aligned} \Pr(L \cap E) &= \Pr(L | E) \cdot \Pr(E) && (\text{law of conditional probability}) \\ &= 0.60 \cdot 0.30 && (\text{substitute the given values}) \\ &= 0.18 && (\text{multiply}) \end{aligned}$$

Theorem 0.5 (Law of total probability). If $B_1, B_2, \dots$ is a *partition* of the sample space, with $\Pr(B_i) > 0$ for every i , then for any event A :

$$\Pr(A) = \sum_i \Pr(A | B_i) \cdot \Pr(B_i)$$

i Proof

Proof. Since $B_1, B_2, \dots$ partition the sample space, the events $A \cap B_1, A \cap B_2, \dots$ are mutually exclusive and their union is A . By countable additivity (Definition 0.9), and then by Theorem 0.4:

$$\begin{aligned} \Pr(A) &= \sum_i \Pr(A \cap B_i) && (\text{countable additivity for partition of } A) \\ &= \sum_i \Pr(A | B_i) \cdot \Pr(B_i) && (\text{law of conditional probability; } \Pr(B_i) > 0) \end{aligned}$$

□

Theorem 0.6 (Bayes' theorem). For any two events A and B with $\Pr(A) > 0$ and $\Pr(B) > 0$:

$$\Pr(A | B) = \frac{\Pr(B | A) \cdot \Pr(A)}{\Pr(B)}$$

i Proof

Proof. By Definition 0.13 and Theorem 0.4:

$$\begin{aligned} \Pr(A | B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\ &= \frac{\Pr(B \cap A)}{\Pr(B)} && \text{(intersection is commutative: } A \cap B = B \cap A) \\ &= \frac{\Pr(B | A) \cdot \Pr(A)}{\Pr(B)} && \text{(law of conditional probability applied to } \Pr(B \cap A)) \end{aligned}$$

□

Example 0.16 (Positive predictive value of a medical test). Suppose a disease test has 99% sensitivity and 99% specificity, and the prevalence of the disease in the population is 7%.

Let D be the event “person has the disease” and $+$ be the event “test is positive”. Then:

- $\Pr(+ | D) = 0.99$ (sensitivity)
- $\Pr(\neg+ | \neg D) = 0.99$ (specificity), so the false positive rate is $\Pr(+ | \neg D) = 1 - 0.99 = 0.01$
- $\Pr(D) = 0.07$ (prevalence)

By Theorem 0.6, with the denominator expanded by the [law of total probability](#) over the partition $\{D, \neg D\}$:

$$\begin{aligned} \Pr(D | +) &= \frac{\Pr(+ | D) \cdot \Pr(D)}{\Pr(+)} && \text{(Bayes' theorem)} \\ &= \frac{\Pr(+ | D) \cdot \Pr(D)}{\Pr(+ | D) \cdot \Pr(D) + \Pr(+ | \neg D) \cdot \Pr(\neg D)} && \text{(law of total probability)} \\ &= \frac{0.99 \cdot 0.07}{0.99 \cdot 0.07 + 0.01 \cdot 0.93} && \text{(substitute the given values)} \\ &= \frac{0.0693}{0.0693 + 0.0093} && \text{(multiply each term in the numerator and denominator)} \\ &= \frac{0.0693}{0.0786} && \text{(add the denominator's two terms)} \\ &\approx 0.88 && \text{(divide)} \end{aligned}$$

Even with a highly accurate test (99% sensitive and 99% specific), only about 88% of people who test positive actually have the disease, because the disease prevalence is relatively low (7%).